

Introduction to Deep Learning: Term Project

Comparing LSTM and Transformer Encoder Models for Text Classification

Team Details

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Initial Project Plan

Dataset

The dataset in use is from Hugging Face: fancyzhx/ag_news only. The decided split is 108,000 train, 12,000 validation, 7,600 official test examples.

Preprocessing

As per the instructions, we will train using 20,000-token vocabulary, with regex tokenizer, and max length of 128. Further we will use padding masks for pooling and attention.

Models

Two models for comparison:

1. Bidirectional LSTM with 2,825,220 parameters
2. Transformer Encoder with 2,973,444 parameters.

Training

For comparison fairness, both models will have the same parameters as possible. At seed 42, we will use optimizer Adam, learning rate $1e-3$, batch size 64, dropout 0.2, and 6 epochs.

Ablation

For ablation tests, we will explore both dataset size variation from 25-100% and sequence length variation, from 128 to 256.

Expected Analysis

For analyzing, we will be taking examples where both models fail and comparing between them, as well as individual failures between both. This will give an idea of the limits and capabilities of both models.